

Natural Language Processing (CPSC 488)

Catalog Course Description

Concepts of natural language, natural language processing, understanding, and generation. Designing and building various language models using computational approaches to solve common natural language problems as well as other real-world problems.

Student Learning Goals

Upon completion of this course, students will be able to:

- Understand and explain the concepts of natural language, natural language processing, understanding and generation, language models, methodologies to build language models using probabilistic and computational approaches such as machine learning and deep learning;
- Design and implement the language models for common natural language problems such as information retrieval and extraction, topic classification, sentiment analysis, machine translation, language generation, dialogue systems, language understanding, and speech recognition with computational approaches;
- Apply various methodologies and techniques to solve other real-world problems of varying complexity.

Credits: 3 units, available for graduate credit

Prerequisite: MATH 170B and 338; CPSC 131 or CPSC 375; or Computer Science or Computer Engineering graduate standing.

Although **prior knowledge of machine learning** is not required, it is strongly recommended, especially, **familiarity** with **artificial neural networks**.

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Office Hours: Mon and Wed 2:15 pm – 4:00 pm and 5:15pm – 5:30 pm
or by ZOOM appointment

Textbook, lecture notes, and programming language

- No textbook is required. Lecture notes and supplemental materials will be available.
- Python programming

Reference book

- D. Jurafsky and J.H. Martin, Speech and Language Processing

Administrative drops

Any student who misses the first two consecutive class meetings may be dropped from this class unless he/she contacts the instructor or the Computer Science department within 24 hours.

Drop policy

A new university withdrawal policy went into effect beginning the spring 1997 semester (UPS 300.016). The policy "stipulates that authorization to withdraw after the second week and prior to the last three weeks of instruction shall be granted for only the most serious reasons. A serious reason is defined as a physical, medical, emotional, or other condition that has the effect of limiting the student's full participation in the class.... The reason must be documented by the student...." The new University Policy also states specifically that "poor academic performance is NOT evidence of a serious reason for withdrawal."

Communication

You have a CSUF-supplied email account, and that is the only way I have of reaching you outside class. Check your email and course page daily for important class announcements and individual messages. I try to respond to emails within two working days, but occasionally may take longer than that. Plan accordingly, especially around deadlines.

Attendance and class participation

You are expected to attend all lectures and actively participate in discussions and activities. Regular attendance is essential for success in this class, and you are responsible for any materials covered during missed lectures.

Students who complete **at least 50% of the pop quizzes or class activities administered without prior announcement** will receive **70%** of the full credit. Additional completion will increase the credit proportionally, up to full credits for participation and attendance. There will be **no make-up** pop quizzes or class activities.

Assignments

There will be multiple assignments/projects. Detailed information about assignments/projects will be announced later. The grade is based strictly on the quality of the work that shows understanding and applicability of the materials, **not based on** the "hard work component." Bonus work or pop quizzes may be given if necessary. However, after the last day of the class, **no extra work** will be given to raise your scores.

Third-party work may not be used in student work without prior instructor consent. Failure to gain and document instructor consent will be construed as willful academic dishonesty. When a third party's work is incorporated into student work after gaining instructor consent, failure to wholly document the work's origin, copyright, and license will be construed as willful academic dishonesty.

Exams

There will be multiple exams, one after a lecture on each major subject is finished to measure individual performance.

Makeup policy: If you are absent from the exam and have a valid and documented excuse, such as an illness, a death in your family, or another equally compelling reason, the weight of your

final will be increased by the weight of the exam. Without a valid and documented excuse, you will receive a zero score for the missed exam, and the final weight will remain unchanged.

Final exam

The final exam will be **comprehensive**. Refer to the university-wide final exam schedule.

Makeup-Policy: If you are absent from the final exam and have a valid and documented excuse, such as an illness, a death in your family, or another equally compelling reason, an incomplete grade will be given. Without a valid and documented excuse, you will receive a zero score for the final exam.

Exam formats and policies

All exams, except pop quizzes, will be announced at least three days in advance. Exam formats may include T/F, multiple choice, short answer, fill-in-the-blank, and other types. **All exams will be closed books and notes and must be completed in person and no AI tools are allowed.**

Additional work and score verification

Bonus work or pop quizzes may be offered as needed for bonus points or to track class attendance. However, **after the last day of the class, no extra work** will be given to raise your scores.

Ensure you verify all your scores before the semester ends.

- The records for assignments, projects, and exams will be posted on the class page. It is your responsibility to review these records and report any discrepancies promptly – no later than **the last day of class**.
- Retain all records of your work, including timestamps, until you receive your final grade.

The university polish for graduate students taking 400-level courses

This course can be used for undergraduate or graduate elective credit.

*According to the university policy (UPS 411.100, section V.C), **graduate students** enrolled in 400-level courses require additional coursework and must demonstrate, in their written and oral performance in the course, a higher quality than that expected of an undergraduate.*

Grading Policy

Your grade will be determined by several of your activities, as shown below.

<i>Class activities</i>	<i>Weights in Percentage</i>
Class participation and attendance	7%
Assignments	33%
Exams	35%
Final exam	25%
Total	100%

* The final grade is computed by weighted sum, $\sum w_i x_i$ where w_i is the weight shown in the above table and x_i is your score on the basis of the 100 percentile. Letter grades will be assigned using the plus/minus scheme as follows ($\geq$ indicates the range minimum):

A: $\geq 92\%$ A-: $\geq 90\%$

B+: $\geq 87\%$ B: $\geq 82\%$ B-: $\geq 80\%$

C+: $\geq 77\%$ C: $\geq 72\%$ C-: $\geq 70\%$

D+: $\geq 67\%$ D: $\geq 62\%$ D-: $\geq 60\%$

*Grades may be assigned by your performance relative to the class average, especially when the class average is low (e.g., < 80). However, no one will be penalized if the class as a whole does excellent work. If given, the weight of bonus work or quiz may be determined by me, considering the class average **within 5%** of the overall score.

AI Use Policy

In this course, the **use of AI tools**, such as Large Language Models (LLMs), **is permitted** for completing assignments/projects, e.g., writing code and rephrasing short answers. However, you should understand any AI-generated code you submit and be able to demonstrate your contribution to the assignment. **No AI tools are allowed during exams** unless otherwise stated, since exams are intended to evaluate *your* understanding of the material.

Academic Dishonesty

Academic dishonesty, such as plagiarism or cheating, will result in a **grade of F** in this course. Repeating offenses may have more severe consequences, up to and including expulsion from the University.

The following is taken from the University Rules (UPS 300,021):

"Academic dishonesty includes such things as cheating, inventing false information or citations, plagiarism, and helping someone else commit an act of academic dishonesty. It usually involves an attempt by a student to show possession of a level of knowledge or skill which he/she in fact, does not possess."

"Cheating is defined as the act of obtaining or attempting to obtain credit for work by the use of any dishonest, deceptive, fraudulent, or unauthorized means. Examples of cheating include, but are not limited to, the following: using notes or aids or the help of other students on tests and examinations in ways other than those expressly permitted by the instructor, plagiarism as defined below, tampering with the grading procedures, and collaborating with others on any assignment where such collaboration is expressly forbidden by an instructor. Violation of this prohibition of collaboration shall be deemed an offense for the person or persons collaborating on the work, in addition to the person submitting the work."

"Plagiarism is defined as the act of taking work of another and offering it as one's own without giving credit to that source. When sources are used in a paper, acknowledgment of the original author or source must be made through appropriate references and, if directly quoted, quotation marks or indentations must be used."

Student information for course syllabus

<https://fdc.fullerton.edu/teaching/student-info-syllabi.html>

Technical Support

In case of technical problems with the learning management system, please contact the help desk at (657) 278-8888.

Disabled Student Services

The University requires students with disabilities to register with the Office of Disabled Student Services (DSS) within the first week of classes. The office of Disability Support Services' website is <http://www.fullerton.edu/DSS/>. They can be reached by phone at (657) 278-3117 or TDD at 657-278-2786. Their email address is dsservices@fullerton.edu. Their office is located in University Hall, room 101. Students requesting accommodations should inform the instructor during the first week of classes about any disability or special needs that may require specific arrangements/accommodations related to attending class sessions, completing course assignments, writing papers or quizzes/tests/examinations."

Emergency Policy (earthquakes, fire, etc.)

For your own safety and the safety of others, each student is expected to read and understand the guidelines

published at «<http://prepare.fullerton.edu/campuspreparedness/>». Should an emergency occur, follow the

instructions given to you by faculty, staff, and public safety officials. An emergency information recording is available by calling the Campus Operation and Emergency Closure line at 657-278-4444.

Tentative Lecture Schedule

A tentative schedule ("**subject to change**") for this class is shown below:

<i>Week</i>	<i>Topics</i>
1–2	Review of Machine Learning and Artificial Neural Networks High-Performance Computing with GPUs, NRP's Nautilus HyperCluster
2–3	An overview of Natural Language Processing (NLP) Text processing, text classification, and sentiment analysis
4–6	Probabilistic language modeling, vector semantics, and embeddings
7	Sequence modeling Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM)
8–9	Transformers and Large Language Models (LLMs) Generative Pretrained Transformer, Finetuning, RAG Language and code generation, Vision Transformer, and Multimodal LLMs
10	Dialogue systems and Question Answering (QA)
11–12	Information Extraction (IE): Sequence labeling and parsing Parts-Of-Speech (POS) and Hidden Markov Model (HMM) for POS tagging Named Entity Recognition (NER)
12–14	Ontologies and knowledge graphs Autonomous intelligent agents, reasoning loop patterns, agentic AI systems
15	Speech recognition and voice synthesis
16	Final exam (see the university-wide schedule)